

## PERTH GOOSEBERRY HILL, Australia

WMO: 946150

Lat: 31.941S

Lon: 116.051E

Elev: 220

StdP: 98.71

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 6.7        | 7.6 | -3.3                          | 2.9 | 21.9 | -1.0 | 3.6 | 20.7 | 13.3                  | 12.0 | 12.1 | 12.0 | 5.5                   | 100  | 0.573 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 13.6                   | 36.9            | 19.7 | 34.9 | 19.4 | 33.1 | 19.1 | 22.8                | 30.8 | 21.7 | 29.7 | 20.8 | 28.8 | 4.8                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.5                            | 15.6 | 25.7 | 19.3 | 14.5 | 24.3 | 18.4 | 13.6 | 23.5 | 68.4          | 30.8 | 64.3 | 29.6 | 60.9 | 29.0 | 27.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 13.2          | 11.6 | 10.4 | 5.0                        | 40.8 | 0.6                | 1.6 | 4.6                                                | 42.0 | 4.2        | 42.9 | 3.9        | 43.8 | 3.5        | 45.0 |     |     |
| (5)               |      |      | 3.7                        | 24.7 | 0.8                | 1.2 | 3.1                                                | 25.5 | 2.6        | 26.2 | 2.1        | 26.8 | 1.5        | 27.7 | (4) | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.4  | 24.3  | 24.6  | 22.7  | 20.0  | 16.4  | 13.9  | 12.7  | 13.2  | 14.3  | 16.9  | 19.8  | 22.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 5.14  | 3.47  | 3.40  | 3.39  | 3.03  | 2.49  | 2.09  | 1.86  | 2.12  | 2.56  | 2.97  | 3.51  | 3.70 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 4     | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 1     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 775   | 1     | 1     | 3     | 15    | 71    | 134   | 175   | 158   | 123   | 66    | 23    | 6    | (9)  |
| (10) |                                                                                  | CDD10.0     | 3066  | 443   | 409   | 393   | 301   | 198   | 117   | 85    | 102   | 131   | 213   | 293   | 382  | (10) |
| (11) |                                                                                  | CDD18.3     | 796   | 186   | 177   | 137   | 66    | 11    | 0     | 0     | 1     | 3     | 21    | 66    | 129  | (11) |
| (12) |                                                                                  | CDH23.3     | 7766  | 2037  | 1876  | 1227  | 393   | 42    | 0     | 0     | 2     | 22    | 176   | 633   | 1359 | (12) |
| (13) | CDH26.7                                                                          | 3487        | 1022  | 924   | 514   | 107   | 4     | 0     | 0     | 0     | 3     | 45    | 243   | 624   | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 5.4   | 6.2   | 6.1   | 5.8   | 5.2   | 5.0   | 4.9   | 5.0   | 5.0   | 5.3   | 5.6   | 5.7   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 764   | 14    | 11    | 24    | 36    | 93    | 140   | 151   | 136   | 87    | 37    | 24    | 11   | (15) |
| (16) |                                                                                  | PrecMax     | 1013  | 107   | 82    | 88    | 108   | 174   | 274   | 250   | 217   | 189   | 89    | 78    | 42   | (16) |
| (17) |                                                                                  | PrecMin     | 438   | 0     | 0     | 0     | 6     | 36    | 29    | 35    | 25    | 21    | 1     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 130   | 25    | 18    | 23    | 28    | 36    | 65    | 53    | 40    | 36    | 23    | 22    | 11   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 39.5  | 39.2  | 37.1  | 32.4  | 26.9  | 21.7  | 20.0  | 22.4  | 25.9  | 31.0  | 35.4  | 38.5 | (19) |
| (20) |                                                                                  |             | MCWB  | 19.7  | 20.0  | 19.0  | 17.6  | 15.2  | 13.1  | 13.1  | 13.9  | 14.6  | 16.8  | 17.6  | 19.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 36.5  | 36.3  | 34.3  | 29.6  | 24.2  | 20.0  | 18.4  | 20.1  | 22.8  | 27.5  | 31.9  | 35.5 | (21) |
| (22) |                                                                                  |             | MCWB  | 20.2  | 20.1  | 19.1  | 17.3  | 14.9  | 13.6  | 12.7  | 13.7  | 14.6  | 16.5  | 17.6  | 19.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 34.3  | 34.3  | 31.6  | 27.4  | 22.3  | 18.5  | 17.1  | 18.2  | 20.6  | 25.1  | 29.3  | 32.6 | (23) |
| (24) |                                                                                  |             | MCWB  | 19.8  | 19.7  | 18.5  | 17.0  | 14.2  | 13.1  | 12.4  | 13.0  | 14.0  | 15.7  | 17.2  | 18.6 | (24) |
| (25) |                                                                                  | 10%         | DB    | 32.2  | 32.2  | 29.4  | 25.5  | 20.5  | 17.3  | 15.9  | 16.9  | 18.8  | 22.8  | 27.1  | 30.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 19.4  | 19.5  | 18.2  | 16.4  | 14.0  | 12.9  | 12.3  | 12.7  | 13.4  | 15.1  | 16.6  | 18.1 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 24.4  | 24.8  | 22.8  | 20.9  | 18.6  | 16.6  | 16.2  | 16.6  | 17.2  | 19.2  | 21.0  | 22.2 | (27) |
| (28) |                                                                                  |             | MCD B | 32.6  | 31.5  | 30.2  | 25.7  | 21.6  | 18.5  | 17.2  | 18.4  | 21.4  | 25.3  | 28.8  | 30.7 | (28) |
| (29) |                                                                                  | 2%          | WB    | 22.7  | 23.0  | 21.5  | 19.8  | 17.5  | 15.7  | 14.8  | 15.4  | 16.2  | 18.0  | 19.6  | 20.9 | (29) |
| (30) |                                                                                  |             | MCD B | 30.8  | 31.4  | 28.8  | 24.5  | 20.0  | 17.4  | 16.0  | 17.3  | 19.9  | 24.5  | 27.6  | 30.4 | (30) |
| (31) |                                                                                  | 5%          | WB    | 21.7  | 21.6  | 20.5  | 18.9  | 16.5  | 14.8  | 13.9  | 14.5  | 15.4  | 17.0  | 18.6  | 20.0 | (31) |
| (32) |                                                                                  |             | MCD B | 29.8  | 29.8  | 27.0  | 23.9  | 19.3  | 16.5  | 15.5  | 16.6  | 18.8  | 22.4  | 25.9  | 29.5 | (32) |
| (33) |                                                                                  | 10%         | WB    | 20.6  | 20.5  | 19.5  | 17.9  | 15.6  | 14.0  | 13.1  | 13.6  | 14.5  | 16.0  | 17.6  | 19.0 | (33) |
| (34) |                                                                                  |             | MCD B | 28.8  | 29.0  | 26.1  | 22.7  | 18.7  | 15.8  | 14.7  | 15.7  | 17.6  | 21.1  | 24.4  | 27.7 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 13.8  | 13.6  | 12.0  | 9.7   | 7.7   | 6.2   | 6.1   | 7.0   | 8.2   | 10.3  | 11.8  | 13.2 | (35) |
| (36) |                                                                                  |             | MCDBR | 16.9  | 16.0  | 14.8  | 12.3  | 10.1  | 7.6   | 7.5   | 8.9   | 11.0  | 14.1  | 15.8  | 16.8 | (36) |
| (37) |                                                                                  |             | MCWBR | 5.9   | 5.5   | 5.3   | 5.0   | 4.9   | 4.3   | 4.5   | 5.0   | 5.6   | 6.5   | 6.4   | 6.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 13.8  | 13.3  | 11.9  | 9.7   | 7.1   | 5.7   | 5.5   | 6.9   | 9.1   | 11.5  | 13.1  | 14.3 | (38) |
| (39) |                                                                                  |             | MCWBR | 5.9   | 5.7   | 5.2   | 4.9   | 4.5   | 4.3   | 4.5   | 4.8   | 5.4   | 5.9   | 6.1   | 6.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.331 | 0.331 | 0.317 | 0.309 | 0.300 | 0.292 | 0.292 | 0.298 | 0.318 | 0.319 | 0.320 | 0.320 | (40) |      |
| (41) |                                                                                  | taud        | 2.584 | 2.599 | 2.633 | 2.641 | 2.637 | 2.641 | 2.639 | 2.614 | 2.531 | 2.551 | 2.571 | 2.593 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 1008  | 989   | 971   | 927   | 886   | 869   | 887   | 925   | 952   | 990   | 1013  | 1023  | (42) |      |
| (43) |                                                                                  | Edh at Noon | 105   | 100   | 91    | 81    | 72    | 68    | 71    | 81    | 99    | 104   | 106   | 105   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 8.19  | 7.37  | 6.09  | 4.45  | 3.32  | 2.72  | 2.88  | 3.73  | 4.97  | 6.47  | 7.72  | 8.38  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.39  | 0.26  | 0.30  | 0.26  | 0.22  | 0.19  | 0.18  | 0.18  | 0.28  | 0.30  | 0.34  | 0.36  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | -0.53 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (19 neighbors) | N/A   | N/A     | N/A    | +0.55   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page